

Chapter 2 Section 1

Examples:

Example 2.1:

Sample space for rolling a die is given as: $S = \{1,2,3,4,5,6\}$

Example 2.2:

Sample space for the given condition is as follows: $S = \{HH, HT, T1, T2, T3, T4, T5, T6\}$

Example 2.3:

According to given condition, there are three items that are selected that may be defective or non-defective, thus, using formula for sample space, we have:

$$O(S) = r^n = 2^3 = 8$$

$$\Rightarrow S = \{DDD, DDN, DND, DNN, NDD, NDN, NND, NNN\}$$

Example 2.6:

Using formula for compliment of a set, we have: $A' = \{\text{cell phone, mp3}\}$

Example 2.8:

$$V \cap U = \{\} \Rightarrow \text{Mutually exclusive}$$

Example 2.10:

$$A \cup B = \{a,b,c,d,e\}$$

Example 2.12:

$$M \cup N = \{z \mid 3 < z < 12\}$$

Excercises:

Excercise 2.1:

- (a) $S = \{8, 16, 24, 32, 40, 48\}$ (b) $S = \{x \mid (x+5)(x-1) = 0\} = \{x \mid x = \{-5, 1\}\}$ or $S = \{-5, 1\}$ (c) $S = \{T, HT, HHT, HHH\}$ (d) $S = \{\text{Asia, Africa, Antartica, Australia, Europe, North America, South America}\}$ (e) $S = \{\}$

Exercise 2.2:

For $r = 3$, we have: $x^2 + y^2 = r^2 \Rightarrow x^2 + y^2 = 9 \Rightarrow S = \{(x,y) | x^2 + y^2 = 9, x \geq 0, y \geq 0\}$

Exercise 2.3:

(a) $A = \{1,3\}$ (b) $B = \{1,2,3,4,5,6\}$ (c) $C = \{x | (x-1)(x-3) = 0\} = \{1,3\}$ (d) $D = \{0,1,2,3,4,5,6\}$

Hence, it is clear that $A = C$ i.e. A and C are equal events

Exercise 2.4:

(a) $S = \{(1,1),(1,2),(1,3),(1,4),(1,5),(1,6),(2,1),(2,2),(2,3),(2,4),(2,5),(2,6),(3,1),(3,2),(3,3),(3,4),(3,5),(3,6),(4,1),(4,2),(4,3),(4,4),(4,5),(4,6),(5,1),(5,2),(5,3),(5,4),(5,5),(5,6),(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$

(b) By rule method, we have: $S = \{(x,y) | 1 \leq x \leq 6, 1 \leq y \leq 6\}$

Exercise 2.5:

Using a tree diagram, we will get the hierarchical order as:

- 1
 - H
 - H
 - T
 - T
 - H
 - T
- 2
 - H
 - T
- 3
 - H
 - H
 - T
 - T
 - H
 - T
- 4
 - H
 - T
- 5
 - H
 - H
 - T
 - T
 - H
 - T
- 6

- H
- T

Hence, we have: $S = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}$

Example 2.6:

Since, two juniors can be selected at a time, hence, the total sample space cannot be given by the sample points formula, instead we have to compute it manually

$$S = \{A_1 A_2, A_1 A_3, A_1 A_4, A_2 A_3, A_2 A_4, A_3 A_4\}$$

Example 2.7:

We have:

$$S_1 = \{MMMM, MMMF, MMFM, MMFF, MFMM, MFMF, MFFM, MFFF, FMMM, FMMF, FMFM, FMFF, FFMM, FFMF, FFFM, FFFF\}$$

$$S_2 = \{0, 1, 2, 3, 4\}$$

Example 2.8:

(a) $A = \{(3,6), (4,5), (4,6), (5,4), (5,5), (5,6), (6,3), (6,4), (6,5), (6,6)\}$ (b) $B = \{(1,2), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,2), (4,2), (5,2), (6,2)\}$ (c) $C = \{(5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$ (d) $A \cup C = \{(5,4), (5,5), (5,6), (6,3), (6,4), (6,5), (6,6)\}$ (e) $A \cap B = \emptyset$ (f) $B \cap C = \{(5,2), (6,2)\}$

Example 2.9:

(a) $A = \{1HH, 1HT, 1TH, 1TT, 2H, 2T\}$ (b) $B = \{1TT, 3TT, 5TT\}$ (c) $A' = \{3HH, 3HT, 3TH, 3TT, 4H, 4T, 5HH, 5HT, 5TH, 5TT, 6H, 6T\}$ (d) $A' \cap B = \{3TT, 5TT\}$ (e) $A \cup B = \{1HH, 1HT, 1TH, 1TT, 2H, 2T, 3TT, 5TT\}$

Excercise 2.10:

(a) A total of three rivers have been taken for fishing, hence, the total outcomes can be given by: $O(S) = r^n = 2^3 = 8$. If F be the event that it is safe to fish and N be the event that it is not safe to fish, then we have:

$$S = \{FFF, FFN, FNF, FNN, NFF, NFN, NNF, NNN\}$$

(b) For two rivers safe for fishing, all events with atleast two F outcomes should be selected, hence, we have:

$$S = \{FFF, FFN, FNF, NFF\}$$

(c) Comparing the outcomes given, we see that each element in the sample set has the middle value as 'F' i.e. the second river is safe would be an event that can define the given set clearly.

Exercise 2.11:

Let the events M_1, M_2, F_1, F_2 be the respective events that represent the selection of the given male and female applicants, then:

- (a) $S = \{M_1 M_2, M_1 F_1, M_1 F_2, M_2 M_1, M_2 F_1, M_2 F_2, F_1 M_1, F_1 M_2, F_1 F_2, F_2 M_1, F_2 M_2, F_2 F_1\}$
- (b) $A = \{M_1 M_2, M_1 F_1, M_1 F_2, M_2 M_1, M_2 F_1, M_2 F_2\}$
- (c) $B = \{M_1 F_1, M_1 F_2, M_2 F_1, M_2 F_2, F_1 M_1, F_1 M_2, F_2 M_1, F_2 M_2\}$
- (d) $C = \{F_1 F_2, F_2 F_1\}$
- (e) $A \cap B = \{M_1 F_1, M_1 F_2, M_2 F_1, M_2 F_2\}$
- (f) $A \cup C = \{M_1 M_2, M_1 F_1, M_1 F_2, M_2 M_1, M_2 F_1, M_2 F_2, F_1 F_2, F_2 F_1\}$

Exercise 2.12:

(a) According to given notations and conditions, we have:

Half of the 3 groups are on salt-free diet whereas other half is not. Moreover, these groups are exercising and performing diet so they are medication free $\Rightarrow \{ZNF, WNF, SNF, ZYF, WYF, SYF\}$

Additional group does not exercise, hence, will belong to the sedentary group. Moreover, they have no restriction on salt and thus, take medication $\Rightarrow \{ZYM\}$

Hence, combining the above, we get the sample space of the given case as:

$$S = \{ZNF, WNF, SNF, ZYF, WYF, SYF, ZYM\}$$

(b) $A = \{ZNF, WNF, SNF, ZYF, WYF, SYF\}$ and $B = \{WNF, WYF\} \Rightarrow A \cup B = \{ZNF, WNF, SNF, ZYF, WYF, SYF\}$

(c) $A \cap B = \{WNF, WYF\}$

Exercise 2.14:

(a) $A \cup C = \{0, 2, 3, 4, 5, 6, 8\}$ (b) $A \cap B = \{\}$ (c) $C' = \{0, 1, 6, 7, 8, 9\}$ (d) $(C' \cap D) \cup B = \{1, 3, 5, 6, 7, 9\}$ (e) $(S \cap C)' = \{0, 1, 6, 7, 8, 9\}$ (f) $A \cap C \cap D' = \{2, 4\}$

Exercise 2.15:

(a) $A' = \{\text{nitrogen, potassium, uranium, oxygen}\}$ (b) $A \cup C = \{\text{copper, sodium, zinc, oxygen}\}$ (c) $(A \cup B) \cup C' = \{\text{copper, sodium, nitrogen, potassium, uranium, zinc}\}$ (d) $B \cap C' = \{\text{copper, uranium, zinc}\}$ (e) $A \cap B \cap C = \{\}$ (f) $(A' \cup B') \cap (A' \cap C) = \{\text{oxygen}\}$

Exercise 2.16:

$$M \cup N = \{x \mid 0 < x < 9\} \quad M \cap N = \{x \mid 1 < x < 5\} \quad M' \cap N' = \{x \mid x = 0, x \geq 9\} \cap \{x \mid x \geq 5\} = \{x \mid 9 \leq x \leq 12\}$$

Exercise 2.18:

(a) The lowest score made by the golfer is why he is losing the tournament, hence, the two events are not mutually exclusive. (b) Getting a flush in one turn does not affect the event of having three of a kind in some other turn, hence, the two events are mutually exclusive. (c) The twin daughters are born on the same day as the baby girl, which means that the baby girl is one of the twins and hence these events are not mutually exclusive. (d) A chess player losing a game does not affect the event that he wins a game in the future, hence, these events are mutually exclusive.

Exercise 2.19:

We will evaluate a particular expression based on the given region, simplify it and then find the actual result

Let M be the event that family experiences mechanical problems Let T be the event that family will receive a ticket for traffic violence Let V be the event that family will arrive at a campsite with no vacancies

Then:

(a) For given region, we have the expression: $M \cap V' \cap T'$ By the evaluated expression, we get the statement: "Family will experience mechanical problems and will not receive a ticket for traffic violence nor will they arrive at a campus with no vacancies".

(b) For given region, we have the expression: $M' \cap V \cap T$ By the evaluated expression, we get the statement: "Family will not experience mechanical problems but will receive a ticket for traffic violation and will arrive at a campsite with no vacancies."

(c) For given region, we have the expression: $(M \cap V \cap T) \cap (M \cap V \cap T') = (M \cap V)$ By the evaluated expression, we get the statement: "Family will experience mechanical problems and will arrive at a campsite with no vacancies."

(d) For given region, we have the expression: $(M \cap V' \cap T) \cap (M' \cap V' \cap T) = (V' \cap T)$ By the evaluated expression, we get the statement: "Family will receive a ticket for traffic violation but will not arrive at a campsite with no vacancies".

(e) For given region, we have the expression: $(M' \cap V \cap T) \cap (M' \cap V \cap T') \cap (M' \cap V' \cap T) \cap (M' \cap V' \cap T') = (M' \cap V) \cap (M' \cap V') = M'$ By the evaluated expression, we get the statement: "Family will not experience mechanical problems."

Exercise 2.20:

We will evaluate a particular expression, and then name the regions that apply to that expression

(a) $M' \cap T' \cap V = 6$ (b) $(M \cap V) \cap T' = 2$ (c) $(M \cup V) \cap T' = 2, 5, 6$ (d) $V' = 4, 5, 7, 8$

Chapter 2 Section 2

Examples:

Example 2.13:

The total sample points for two consecutive events can be given using the multiplication rule of counting i.e. we have: $O(S) = n_1 * n_2 = 6 * 6 = 36$ points

The other method to find this is using the traditional sample space formula i.e. $O(S) = r^n$. Using this formula we would get $O(S) = 6^2 = 36$ points

Example 2.14:

The ways in which a buyer can order one of the homes is clearly based upon the different combinations yielded by exterior styling and floor plans. In such situations, we apply the multiplication rule of counting i.e. we have: $O(S) = n_1 * n_2 = 4 * 3 = 12$ (Since there are 4 exterior stylings and 3 floor plans)

Example 2.15:

In this case, we have to elect 2 people from 22 such that one person cannot be the chair and treasurer at the same time. Hence, we can select the first out of 22 and the second out of remaining 21. Using the multiplication rule for counting, we have: $O(S) = n_1 * n_2 = 22 * 21 = 462$ ways

Example 2.16:

This case is the same as the previous example 2.14 but with more items included. Sam can build the computer using the combination of different items he has. Applying the multiplication rule, we have: $O(S) = n_1 * n_2 * n_3 * n_4 = 2 * 4 * 3 * 5 = 120$ ways

Example 2.17:

In this problem, we can have two cases for four digit even numbers i.e.

(a) The last digit is zero (b) The last digit is 2 or 6

(a) In case the last digit is zero, then we have 1 possibility for unit number, 5 possibilities for tens, 4 possibilities for hundred and 3 possibilities for thousand i.e. $O(S) = n_1 * n_2 * n_3 * n_4 = 1 * 5 * 4 * 3 = 60$ possibilities

(b) In case the last digit is not zero, then we have 2 possibilities for unit number (either 2 or 6), 4 possibilities for thousand (since 0 cannot be used in thousand place otherwise number will become 3 digit), 4 possibilities for hundred place and 3 possibilities for tens place, i.e. $O(S) = n_1 * n_2 * n_3 * n_4 = 2 * 4 * 4 * 3 = 96$ possibilities

Combining these two, we will get the final result i.e. $O(S) = 60 + 96 = 156$ possibilities

Note: In these type of questions, where conditions are specified, we prioritize conditions over the default order of possibilities e.g. one might have for (b) part consider putting 2 possibilities in unit place (due to 2 or 6) and then putting 5 in place of tens place.. but we didn't do that

because we had another condition that 0 could not be used in the thousands place. SO we first considered the thousand place possibilities and then moved to hundred. Notice that since the conditions were only two, thus, switching the possibilities between the hundred and tens place would not change the result in anyway i.e. if we put $n_3 = 3$ and $n_4 = 4$, it does not change the final result.

Example 2.18:

The multiplication rule of counting can be summarized to a permutation formula if and only if the order for the items does not matter and repetition is not allowed, otherwise, always use the multiplication formula. Here, it is being told that 3 awards are to be given to 25 graduate student with each student getting only one award at a time (i.e. repetition is not allowed and it does not matter whether the 5th student gets the award or the 22nd), hence, we can apply formula for permutation for the given event as: $O(A) = {}^n P_r = {}^{25} P_3 = 25!/(25-3)! = 25!/22! = (25)(24)(23) = 13,800$ possible selections (Same as applying multiplication rule of counting but simpler to evaluate)

Example 2.19:

(a) For the first case, there are no restrictions i.e. anyone can become a president and treasurer, hence, applying formula for permutation: $O(S) = {}^n P_r = {}^{50} P_2 = 2450$ choices

(b) For the second case, it is the condition that A will only serve as officer if he is the president otherwise not. Hence we can have two sub-cases here i.e.

(i) A is the president: then the remaining 49 can become the treasurer, thus we have: $O(A_1) = 49$ (ii) A is not the president: then he will not serve as an officer and the total number of officers reduce from 50 to 49. Now from these 49 there are no restrictions, hence applying formula for permutation: $O(A_2) = {}^n P_r = {}^{49} P_2 = 2352$

Thus, total choices for this case become $O(A) = 2352 + 49 = 2401$ choices

(c) For the third case, it is the condition that B and C will either serve together or not at all. Hence, we can have two sub-case here i.e.

(i) B and C both serve: then they both occupy the position of the leader and treasurer i.e. $O(B_1) = 2$ (ii) B and C both do not serve: then the total number of officers to select from will be 48. Hence, we can apply the permutation formula on the remaining officers as: $O(B_2) = {}^n P_r = {}^{48} P_2 = 2256$

Thus, the total choices for the third case become $O(B) = 2 + 2256 = 2258$ choices

(d) For the fourth case, it is given that D and E will not serve together, this can be computed by first computing the possibility that when D and E will serve together and take the compliment of that event. Hence, we have for D and E serving together: $O(C) = 2$ (as both will take either treasurer or leader's place). Now we have, $O(C') = O(S) - O(C) = 2450 - 2 = 2448$ choices

Example 2.20:

The given example tells us to apply permutations based on the classes of the students. Hence, we will apply the permutation formula for repeated items i.e. $O(A) =$

$$n!/(n_1! * n_2! * n_3! * \dots * n_k!) = 10!/(1! * 2! * 4! * 3!) = 12,600 \text{ ways}$$

Note: Permutation formula for repeated items is applied if and only if the item belonging to a particular kind or class is repeating again.

Example 2.21:

In this case, we are partitioning a group of 7 students (not of different kinds, same kind being distributed on some basis) into 1 triplet and 2 double hotel rooms. Although the formula for both partition and repetition is same, it must be known that partition is always when you distribute a given class on some basis whereas repetition is when we have to find permutation on repeated values. Applying formula for partition, we have: $O(A) = n!/(n_1! * n_2! * n_3!) = 7!/(3! * 2! * 2!) = 210 \text{ ways}$. (since each triplet would mean that 3 values are same whereas double means 2 values are same)

Example 2.22:

In this case, we are finding ways in which mother can select games and cartridges for her child. In case of Selections, we apply combinations. Hence, the given condition can be distributed into two parts:

(a) Ways of selecting arcade games: $n_1 = {}^nC_r = {}^{10}C_3 = 120 \text{ ways}$ (b) Ways of selecting cartridges: $n_2 = {}^nC_r = {}^5C_2 = 10 \text{ ways}$

Now the total ways of selecting can be given by all possibilities of arcade games with cartridges i.e. we apply multiplication rule: $O(A) = n_1 * n_2 = 120 * 10 = 1200 \text{ ways}$

Example 2.23:

Here again we are given a single word, but we are partitioning it on the basis of repeated words as the given condition for the question is to find the different arrangements of letters in the given word. Hence, we have applying formula for permutation for partition: $O(A) = n!/(n_1! * n_2! * \dots * n_k!) = 10!/(3! * 3! * 2!) = 50,400 \text{ arrangements}$.

Exercises:**Exercise 2.21:**

We are asked the total number of ways in which person can go for sightseeing. We can apply multiplication rule as we need the total number of combinations that can be yielded for given condition: $O(A) = n_1 * n_2 = 6 * 3 = 18 \text{ ways}$

Exercise 2.22:

The total ways of the classification are the total combination that can be yielded for given conditions i.e. 8 blood groups or 3 blood pressure rates: $O(A) = n_1 * n_2 = 8 * 3 = 24$ ways

Exercise 2.23:

The total sample points for this case can be given by using the multiplication rule as the two events are independent of each other and that we need the total outcomes yielded from the combination of these events: $O(A) = n_1 * n_2 = 6 * 26 = 156$ points.

Exercise 2.24:

The total number of possible classifications can be computed on the basis of combining the college year and gender as: $O(A) = n_1 * n_2 = 4 * 2 = 8$ classifications

Exercise 2.25:

The different pairs he will have on display can be given by combining the different style of shoes with the distinct colors i.e. we have: $O(A) = n_1 * n_2 = 5 * 4 = 20$ pairs

Exercise 2.26:

In this case, 7 cases are given and they are telling us as in how many different ways can the people with given conditions adopt 5 of them:

(a) If all 7 rules are being violated then they simple have to adopt 5 of them without any restrictions i.e. we have $O(A) = {}^nP_r = {}^7P_5 = 21$ ways

(b) According to given condition, the people are already following 2 conditions which means that they have to adopt 3 more without any restrictions. Since, the total conditions to adopt have reduced to 5 from 7, hence we have permutation formula as: $O(B) = {}^nP_r = {}^5P_3 = 10$ ways

Exercise 2.27:

The total plans available can be given using the multiplication rule i.e. $O(A) = n_1 * n_2 * n_3 * n_4 = 4 * 3 * 2 * 2 = 48$ plans.

Exercise 2.28:

The different ways in which doctor can prescribe will be given on the basis of all possible outcomes of the manufacturers, form of drug and strength i.e. $O(A) = n_1 * n_2 * n_3 = 5 * 3 * 2 = 30$ ways

Exercise 2.29:

The total test runs can be computed using the multiplication rule as: $O(A) = n_1 * n_2 * n_3 * n_4 = 3 * 5 * 7 * 2 = 210$ test runs.

Exercise 2.30:

In cases like these, for every true/false question, every answer can be given as a true or false value i.e. we have 2 choices. Considering the combinations among the true/false questions as well as between the choices true and false, we have permutation for different answers for each question as $n = 2$. Hence, total combination can be given by multiplication rule on all 9 questions i.e. $O(A) = 2^9 = 512$ ways

Exercise 2.31:

It is given that RLH were the first three letters of the license number (we know that there are total of 6 numbers on license plate) and it is given the first of the three digits of 5. Moreover, according to given condition repetition is not allowed and order doesn't matter, hence we can apply permutation formula. We have for the last two digits: $O(A) = {}^n P_r = {}^9 P_2 = 72$ automobile registrations (We used 9 as there are a total 10 numbers between 0 and 9 inclusive out of which 5 cannot be used as it was the first digit).

Exercise 2.32:

(a) The possible ways for people to line up for a bus does not have any restrictions or order-strictness, hence, we can apply permutation formula i.e. $O(S) = {}^n P_r = {}^6 P_6 = 6! = 720$ ways

(b) The possible ways for people to line up with 3 people following each other can be divided into two sub-processes:

(i) Since 3 of them are following each other, the people that are following without repetition of the following are 4, thus we have applying permutation formula on non-repeated lineups as: $n_1 = n! = 4! = 24$ ways (ii) For the total number of possible combinations, it is also needed that we find the ways of arrangement of the people following each other. Applying permutation formula, we get: $n_2 = n! = 3! = 6$ ways

The total ways can now be given by using the multiplication rule on the total ways as: $O(A) = n_1 * n_2 = 24 * 6 = 144$ ways

(c) The possible ways that two people do not follow each other in a line can be found using compliment operation on the case that two people are following each other. Using the same methodology as in part(b) we get: $O(B) = n_1 * n_2 = 5! * 2! = 120 * 2 = 240$ ways. Now, applying compliment rule, we have for given case: $O(B') = O(S) - O(B) = 720 - 240 = 480$ ways

Exercise 2.33:

This is the same case as in exercise 2.30 but with a different condition and values.

(a) An answer to a question can be check using any of the 4 options (whether it be correct or wrong doesn't matter). Hence, for 5 questions we have by multiplication rule: $O(A) = 4^5 = 1024$ ways

(b) An answer can be wrong if any of the 3 wrong options are checked for a given question. Hence, for 5 questions, we have by multiplication rule: $O(B) = 3^5 = 243$ ways

Exercise 2.34:

(a) The distinct permutations in the given word can be given directly as in this case the given word does not have any repeating letters. Thus, we have: $O(A) = n! = 7! = 5040$ ways

(b) The permutations with the first letter being M can be given the same way. If we fix the letter M at the first place, then only 6 letters remain which we can arrange at will. Hence, using the same formula, we get: $O(B) = n! = 6! = 720$ ways

Exercise 2.35:

In this case, even though it seems that we have to build 6 houses on 1 street and 3 on the other side, but the order doesn't matter here and repetition is not allowed. Hence, we can give the total number of ways he can place houses on the street as: $O(A) = n! = 9! = 362,880$ ways

Exercise 2.36:

(a) Same as in the example 2.17, we have for 3 digit numbers: The hundred position can be occupied by 6 numbers, the tens position by 6 numbers and the unit position by 5 numbers i.e. by multiplication rule: $O(A) = n_1 * n_2 * n_3 = 6 * 6 * 5 = 180$ numbers

(b) In case of odd-digit numbers, we have: The unit position can be occupied by 3 numbers (either 1, 3 or 5), the hundred position can be covered by any 5 (excluding 0) and the tens position can be occupied by any 5, thus, we have: $O(B) = n_1 * n_2 * n_3 = 3 * 5 * 5 = 75$ numbers

(c) In case of the number greater than 330, there can be two cases i.e.

(i) Hundredth position is covered by 3: Here 3 covers the hundredth position, any of the 3 (4, 5 or 6) can cover the tens position (as number has to be greater than 330, we cannot use 0, 1 or 2) and any of the remaining 5 can cover the unit position, i.e. we have: $n_1 = 1 * 3 * 5 = 15$ numbers

(ii) Hundredth position is covered by number greater than 3: Here any of the 3 (4, 5 or 6) can cover the hundredth position, remaining 6 can cover the tens position and last 5 can cover the unit position i.e. we have: $n_2 = 3 * 6 * 5 = 90$ numbers

The total numbers can now be given as: $O(C) = n_1 + n_2 = 15 + 90 = 105$ numbers

Exercise 2.37:

It is given that boys and girls must sit alternately, so we can say that one of 5 girls sits first, then one of 4 boys, then one of remaining 4 girls then remaining of 3 boys and so on. Using the multiplication rule, we have: $O(A) = 5 * 4 * 4 * 3 * 3 * 2 * 2 * 1 * 1 = 2880$ ways

Exercise 2.38:

Four married couples have bought 8 seats is the given condition. Now we have:

(a) With no restrictions means that anybody of the 8 people can sit with anybody i.e. we have using permutation rule: $O(A) = n! = 8! = 40320$ ways

(b) With the condition that each couple sits together, we have four couples i.e. $4!$ number of arrangements with the fact that each couple has its own arrangement of the two people i.e. $2!$ per couple. Thus, we get using the multiplication rule: $O(B) = (2!)^4 * 4! = 384$ ways (We used power 4 over $2!$ as each couple could have an arrangement of $2!$ among themselves. multiplying the result for given 4, we get $(2!)^4$).

(c) If all the men sit right next to all the women, then each men and women group can be arranged in $4!$ ways using the permutation rule. Hence, by multiplication rule, we have: $O(C) = (4!) * (4!) = 576$ ways

Exercise 2.39:

(a) For all 8 finalists, the sample points can be given by the permutation rules for all values i.e. $O(A) = n! = 8! = 40320$ ways

(b) For the first three positions out of 8 finalists, the order doesn't matter and repetition is not allowed, hence, we can use permutation here as well i.e. $O(B) = {}^nP_r = {}^8P_3 = 336$ ways

Exercise 2.40:

The 5 starting positions can be taken without concern regarding order of the players. Moreover, repetition is not allowed here, hence, by using permutation we have: $O(A) = {}^nP_r = {}^8P_5 = 6720$ ways

Exercise 2.41:

The ways in which teachers can be assigned is independent of the order in which they are assigned. Moreover, repetition is not allowed as each teacher can be assigned to only one subject at a time. We use permutation rule to get: $O(A) = {}^nP_r = {}^6P_4 = 360$ ways

Exercise 2.42:

The sample points for the given condition will be independent of the order in which the three lottery tickets are drawn. Since each contestant can hold only one ticket hence repetition is not allowed. By using permutation rule, we have: $O(A) = {}^nP_r = {}^{40}P_3 = 59,280$ ways.

Exercise 2.43:

For the different ways for circular arrangement of trees, we apply the permutation rule for circular arrangements i.e. $O(A) = (n-1)! = (5-1)! = 4! = 24$ ways

Exercise 2.44:

For the to be arranged in the circle, the total number of ways can be given by using circular permutation formula i.e. $O(A) = (n-1)! = (8-1)! = 7! = 5040$ ways

Exercise 2.45:

In this case, we are being asked the distinct permutations of a word in which certain letters are repeating. Hence, we will apply the partition formula for permutation to find the total number of ways i.e. $O(A) = n!/(n_1! * n_2! * \dots * n_k!) = 8!/(3! * 2!) = 3360$ ways.

Exercise 2.46:

In this case, the arrangement is being made on different classes of trees, hence we will use formula for permutation in case of repetition of the same class, i.e. we have: $O(A) = n!/(n_1! * n_2! * \dots * n_k!) = 9!/(3! * 2! * 4!) = 1260$ ways

Chapter 2 Section 3

Examples:

Example 2.24:

The total outcomes for the event are 4 i.e. {HH, HT, TH, TT}. The chances of getting atleast 1 head means that there can be 1 or more than 1 heads in the event, which are 3 i.e. {HH, HT, TH}. Thus, probability for this event can be computed as:

$$P(A) = O(A)/O(S) = 3/4$$

Example 2.25:

For the given condition, we have: $O(\text{even}) = 2 * O(\text{odd})$. Thus, using the given condition, we have: $O(\text{odd number}) = 1$; $O(\text{even number}) = 2$; Total outcomes = $O(S) = 3(1) + 3(2) = 9$ (Since there are 3 odd numbers and 3 even numbers in a die roll). Now, we have for the given event E: $P(E) = P(1) + P(2) + P(3) = 1/9 + 2/9 + 1/9 = 4/9$ (The case stated that any case in which number is less than 4 i.e. we have an OR condition: in case of which we add probabilities of individual events to get the total probability)

Example 2.26:

Let A be the event that an even number turns up in the die roll = {2,4,6} Let B be the event that a number divisible by 3 occurs in the die roll = {3,6} Now,

- $A \cup B = \{2,3,4,6\} \Rightarrow P(A \cup B) = 2/9 + 1/9 + 2/9 + 2/9 = 7/9$
- $A \cap B = \{6\} \Rightarrow P(A \cap B) = 2/9$

Example 2.27:

Let I, M, E and C represent the respective events that the engineer is selected from industrial, mechanical, electrical and civil. Now, we have: $O(S) = 25 + 10 + 10 + 8 = 53$ outcomes

(a) Since we want an industrial engineer major, thus, we have: $P(I) = O(I)/O(S) = 25/53$

(b) Since we want an electrical or civil engineer major, thus, we have: $P(C \text{ or } E) = P(C) + P(E)$
 $= 8/53 + 10/53 = 18/53$

Example 2.28:

Let A be the event that an ace is selected, B be the event that a jack is selected and N be the event of obtaining a poker card.

In case of event A, for dealing 2 aces out of 4 from a deck, we have the outcomes given by the combination formula as we can select any 2 aces from the given 4: $O(A) = {}^nC_r = {}^4C_2 = 6$.

In case of event B, for dealing 3 jacks out of 4 from a deck, we have the outcomes given by the combination formula as we can select any 3 jacks from the given 4: $O(B) = {}^nC_r = {}^4C_3 = 4$

Now since we want both the event A and event B to happen, we have: $O(A \text{ and } B) = O(A) * O(B) = 6 * 4 = 24$ bands

Now lets move to the poker hand event N, the total ways of selecting 5 cards from a eck of 52 cards is also given by combination formula as: $O(N) = {}^nC_r = {}^{52}C_5 = 2,598,960$ ways

Thus, we now have for our required probability of event: $P(A \text{ and } B) = O(A \text{ and } B) / O(S) = 24/2,598,960 = 0.9 * 10^{-5}$

Example 2.29:

Let A be event he gets offer from company A and B be the event that he gets offer from company B, then for given conditions, we have:

$$P(A) = 0.8, P(B) = 0.6, P(A \cap B) = 0.5$$

For the required event, we have using addition rule of probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.6 - 0.5 = 0.9$$

Example 2.30:

Let A be the event that we get a 7 and B be the event that we get an 11, then for the given fair dice, we have:

$$P(A) = P\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\} = 1/36 + 1/36 + 1/36 + 1/36 + 1/36 + 1/36 = 1/6$$

$$P(B) = P\{(5,6),(6,5)\} = 1/36 + 1/36 = 1/18$$

Now, for required probability, applying the additional rule for probability: (Events A and B are mutually exclusive)

$$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) = 1/6 + 1/18 = 2/9$$

Example 2.31:

Let G, W, R and B be the events that the buyer selects a green, white, red and blue automobile respectively. Then we have for the given condition, using addition rule of probability:

$$P(G \cup W \cup R \cup B) = P(G) + P(W) + P(R) + P(B) = 0.09 + 0.15 + 0.21 + 0.23 = 0.68$$

Example 2.32:

Let E be the event that he services atleast 5 cars then we have according to given condition:

$P(E) = 1 - P(\bar{E}) = 1 - P(3) + P(4) = 1 - 0.12 - 0.19 = 0.69$ (Here we used the compliment method for our ease only, the alternative approach would have been to sum up all the probabilities for events having 5 or more cars. When the actual problem is difficult to compute and is very large, it alternatively means that its compliment will be small as in this case, hence, we can find the probability of compliment event first and then use the compliment method to compute the actual probability of the event)

Example 2.33:

Let S, M and L represent the events that the cable manufactured is smaller than requirements, meets the requirements and larger than the requirements respectively then:

(a) $P(L) = 1 - P(S) - P(M) \Rightarrow P(L) = 1 - P(L) + P(M) \Rightarrow 2P(L) = 1 - 0.99 \Rightarrow P(L) = 0.01/2 = 0.005$

(b) The probability that the cable will be larger than 1990 mm is the same as finding the probability that cable will meet requirements or will be larger than them i.e. $P(X \geq 1990) = P(M) + P(L) = 0.99 + 0.005 = 0.995$

Exercises

Excercise 2.49:

(a) Summing the probabilities we get: $P(S) = 0.19 + 0.38 + 0.29 + 0.15 = 1.11 > 1$ (which is not possible) (b) The given events are compliments of each other, but the sum of the probabilities is: $0.40 + 0.52 = 0.92 \neq 1$ (which is against the compliment rule of probability) (c) In the given probabilities, $P(2)$ is negative value which is not possible by the axiom of probability i.e. $0 \leq P(A) \leq 1$. (d) We have been given that the probability of both heart and black card is $1/8$, but it should be known that the Heart cards cannot be black in nature i.e. it must be 0.

Excercise 2.50:

We have for data in example 2.8:

(a) $P(A) = 10/36 = 5/18$ (b) $P(C) = 12/36 = 1/3$ (c) $P(A \cap C) = 7/36$

Excercise 2.51:

Let A, B and C be the respective events that the drawn envelope contains \$100, \$75 and \$25. For sample space for different amounts of money, we have:

$$S_{money} = \{100, 75, 25\}$$

We can obtain the total outcomes by using the number of envelopes for each of these amounts i.e.

$$O(S) = O(A) + O(B) + O(C) = 75 + 150 + 275 = 500$$

Now, for the event that envelope contains less than \$100 , we have:

$P(A') = 1 - P(A) = 1 - 75/500 = 425/500 = 17/20$ (A' event means less than 100 since we only have three events and all of them represent less amount than 100)

Exercise 2.52:

Let A be the event that the student smoke, B be the event that the student drinks alcohol and C be the event that the student eats between meals, then according to given condition, we have:

$$P(A) = 210/500 ; P(B) = 258/500 ; P(C) = 216/500 ; P(A \cap B) = 122/500 ; P(B \cap C) = 83/500 ; P(A \cap C) = 97/500 ; P(A \cap B \cap C) = 52/500$$

(a) We have to compute the probability of the event if student smokes but does not drink alcohol i.e. $P(A \cap B')$. We know that $P(A \cap B)$ gives us the values that are common between two events A and B. Now, if we remove these common values from the values in set A, we will obtain all such values that are common in A and B' (as they are not common between A and B: its obviously a compliment), thus, using this approach, we can find the probability required as:

$$P(A \cap B') = P(A) - P(A \cap B) = 210/500 - 122/500 = 88/500 = 22/125$$

(b) We have to compute the probability of the event if student drinks beverages and eats between meals but does not smoke i.e. $P(A' \cap B \cap C)$. We know that $P(B \cap C)$ gives us the probability of values that are common in both B and C. Now if we remove the value that are common between all three $P(A \cap B \cap C)$ from this value, we will get the probability of all those values that are common in B and C and in A' (as we have removed common values in A, B and C). We have for required probability:

$$P(A' \cap B \cap C) = P(B \cap C) - P(A \cap B \cap C) = 83/500 - 52/500 = 31/500$$

(c) We have to compute the probability of the event that the student neither smokes nor eats between meals i.e. $P(A' \cap C')$. We know that by DeMorgan's law: $(A' \cap C')' = A \cup C$, hence, we need to find the value for $P(A \cup C)$, which can be found using the addition rule:

$$P((A' \cap C')') = P(A \cup C) = P(A) + P(C) - P(A \cap C) = 210/500 + 216/500 - 97/500 = 329/500$$

Now using compliment rule to find the probability for $(A' \cap C')$:

$$P(A' \cap C') = 1 - P((A' \cap C')') = 1 - 329/500 = 171/500$$

Exercise 2.53:

Let A and B be the respective events that the industry is located in Shengai and Beijing, then according to given condition:

$$P(A) = 0.7 ; P(B) = 0.4 ; P(A \cup B) = 0.8$$

(a) For industry to be in both cities, we have to apply the addition rule as: (AND means the operation $\cap$) $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.7 + 0.4 - 0.8 = 0.3$

(b) For industry to be in neither cities, it does not have to be in either of these (either/OR means the operation $\cup$), thus, we can use compliment rule of probability to the value $P(A \cup B)$:

$$P(C) = 1 - P(A \cup B) = 1 - 0.8 = 0.2$$

Exercise 2.54:

Let A and B be the events that customer invests in tax-free bonds and mutual funds respectively, then according to given condition:

$$P(A) = 0.6 ; P(B) = 0.3 ; P(A \cap B) = 0.15$$

(a) Applying the addition rule of probability for mutually inclusive events, we have for either one of them (Either mean the operation $\cup$):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.3 - 0.15 = 0.75$$

(b) For neither of them, we need to find the compliment of the above computed value i.e.:

$$P((A \cup B)') = 1 - P(A \cup B) = 1 - 0.75 = 0.25$$

Exercise 2.55:

We are given that each coded item has 3 distinct letters and then 4 non-zero distinct digits. For such a case, we have the total number of outcomes, using the multiplication rule as:

$$O(S) = (26)(25)(24)(9)(8)(7)(6) = 47,174,400 \text{ coded items}$$

Let E be the event that the first letter of item is vowel and the digits are an even number, then for the given conditions, we can divide it into two parts:

(i) For the first letter out of the three to be a vowel, we have 5 possibilities, followed by 25 and 24 possibilities for the next two letters from the remaining letters. Hence, using multiplication rule, we have: $n_1 = (5)(25)(24) = 3000$ ways

(ii) For the digits to be even in nature, it is necessary that the last digit is an even number i.e. we have 4 possibilities (2,4,6 and 8. Remember we cannot include 0 as the condition says that coded item contain only non-zero digits), with 8,7 and 6 possibilities for the other three digits respectively. Using multiplication rule, we have: $n_2 = (4)(8)(7)(6) = 1344$ ways

Thus, total outcomes for the given event E are: $O(E) = n_1 * n_2 = 3000 * 1344 = 40,320,00$ coded items

Now, using the probability formula, we can get the probability for the event E as: $P(E) = O(E)/O(S) = 40,320,00/47,174,400 = 10/117 = 0.085$

Exercise 2.56:

Let A,B,C,D be the events that there is a defect in brake system, transmission, fuel system and some other area respectively, then for the given condition, we have:

$$P(A) = 0.25 ; P(B) = 0.18 ; P(C) = 0.17 ; P(D) = 0.40$$

(a) We have been given the probability for the defect in both both brakes and fuel system at a time i.e. $P(A \cap C) = 0.15$. Using this we have for the probability for defect in either of the systems (using addition rule) as:

$$P(A \cup C) = P(A) + P(C) - P(A \cap C) = 0.25 + 0.17 - 0.15 = 0.27$$

(b) For no defects in either the brakes or fuel system, we can apply complimentary rule to the above computed value (as this event is the compliment of defect in either of these systems):

$$P((A \cup C)^c) = 1 - P(A \cup C) = 1 - 0.27 = 0.73$$

Exercise 2.57:

(a) There are 5 vowels in English exclusive of y, so the probability would be: $P(A) = 5/26$ (b) "j" is the 10th letter of the English language, hence, there are 9 letters ahead of (come before the letter does) "j" and thus the probability would be: $P(B) = 9/26$ (c) "g" is the 7th letter of the English language, hence, there are 19 more letters after "g" and thus the probability would be: $P(C) = 19/26$

Exercise 2.58:

For a fair dice, the total outcomes are 36. Now:

(a) For a total of sum 8 for the two dices, the values should be in the set $A = \{(2,6),(3,5),(4,4),(5,3),(6,2)\} \Rightarrow O(A) = 5$. Now, for the probability of this event: $P(A) = O(A)/O(S) = 5/36$

(b) For atmost a total of sum 5 for the two dices, the values should be in the set $B = \{(1,1),(1,2),(1,3),(1,4),(2,1),(2,2),(2,3),(3,1),(3,2),(4,1)\} \Rightarrow O(B) = 10$. Now, for the probability of this event: $P(B) = O(B)/O(S) = 10/36 = 5/18$

Exercise 2.59:

For a poker hand consisting of 5 cards, the total number of outcomes will be equal to the selection of 5 cards from a deck of 52, which we can obtain using the formula for combinations, as:

$$O(S) = {}^nC_r = {}^{52}C_5 = 2,598,960 \text{ outcomes}$$

(a) In this case, we are given that 3 of the 5 cards will be aces whereas the other 2 can be any cards. We can divide this problem into two parts;

(i) Compute the outcomes for having 3 aces, which can be done by finding the total outcomes of selecting 3 aces from 4 in a deck. Thus, using combination formula, we have: $n_1 = {}^nC_r = {}^4C_3 = 4$

(ii) Compute the outcomes for having any 2 other cards, which can be done by finding the total outcomes of selecting 2 cards from a deck of 48 cards (48 because 4 were aces and these 2 cannot be an ace). Thus, using combination formula, we have: $n_2 = {}^nC_r = {}^{48}C_2 = 1128$

Thus, total outcomes for the given event are: $O(A) = n_1 * n_2 = 4 * 1128 = 4512$. Now, for the probability of this event: $P(A) = O(A)/O(S) = 4512/2,598,960 = 94/54145$

(b) In this case, we are given that 4 of them would be hearts and 1 of them would be a club. We can divide this problem into two parts:

(i) Compute the outcomes for having 4 hearts, which can be given by finding the total outcomes for selecting 4 hearts from 13 in a deck. Thus, using the combination formula, we have: $n_1 = {}^nC_r = {}^{13}C_4 = 715$

(ii) Compute the outcomes for having 1 club, which can be given by finding the total outcomes for selecting 1 club from 13 in a deck. Thus, using the combination formula, we have: $n_2 = {}^nC_r = {}^{13}C_1 = 13$

Thus, total outcomes for the given event are: $O(B) = n_1 * n_2 = 715 * 13 = 9295$ Now, for the probability of this event: $P(B) = O(B)/O(S) = 9295/2,598,960 = 143/39984$

Exercise 2.60:

For the given case, the total number of outcomes for selecting 3 books can be given by applying the combination formula for 3 books out of 9 present on the shelf, thus, we have: $O(S) = {}^nC_r = {}^9C_3 = 84$

(a) In this case, we are given that 1 dictionary is selected, whereas the other 2 books can be from the remaining. We know that there is only a single dictionary book, therefore, for the selection of a dictionary, we have: $n_1 = 1$. Now, for selecting 2 books from the remaining 8, we can again apply the combination rule to obtain: $n_2 = {}^nC_r = {}^8C_2 = 28$.

Thus, the total number of outcomes for this event are: $O(A) = n_1 * n_2 = 1 * 28 = 28$. Now for the probability: $P(A) = O(A)/O(S) = 28/84 = 1/3$

(b) In this case, we are given that 2 novels and 1 books of poems is selected. This problem can be divided into two parts:

(i) Compute the outcomes for having 2 novels, which can be given by finding the total outcomes for selecting 2 novels out of 5 on the shelf. Thus, using the combination formula, we have: $n_1 = {}^nC_r = {}^5C_2 = 10$.

(ii) Compute the outcomes for having 1 book of poems, which can be given by finding the total outcomes for selecting 1 poem book out of 3 on the shelf. Thus, using the combination formula, we have: $n_2 = {}^nC_r = {}^3C_1 = 3$

Thus, total outcomes for the given event are: $O(B) = n_1 * n_2 = 10 * 3 = 30$ Now, for probability of the event: $P(B) = O(B)/O(S) = 30/84 = 5/14$

Exercise 2.61:

Let A,B be the events that student has studied Mathematics and History respectively, then according to given condition:

$$P(A) = 54/100 ; P(B) = 69/100 ; P(A \cap B) = 35/100$$

(a) For a student to have taken mathematics or history, we need to find the probability of the event $(A \cup B)$, which can be found using the addition rule as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 54/100 + 69/100 - 35/100 = 88/100$$

(b) For a student to have not taken either of the subject, we can apply the complimentary rule on the above computed value as:

$$P((A \cup B)') = 1 - P(A \cup B) = 1 - 0.88 = 0.12$$

(c) For a student to take history but not mathematics means that we want the values for history only i.e. $P(B)$, removing the values common between both subjects i.e. $P(A \cap B)$, thus, we have for this new event C:

$$P(C) = P(B) - P(A \cap B) = 69/100 - 35/100 = 34/100 = 17/50$$

Exercise 2.62:

(a) For the given case, the sample space of the scenario will tell the total number of combinations involved i.e we have 3 possibilities for each which means that using the multiplication rule, we get: $O(S) = 9$

(b) For the case that judge gets a thin crust with standard sauce, we have this as a single outcome in our sample space, hence, $O(A) = 1$. Using probability rule, we have: $P(A) = O(A)/O(S) = 1/9$. (The given case is referred to as a single outcome because we have for the sample space outcomes: {"Thin with standard", "Thin with a new sauce and more garlic", "Thin and a new sauce with mroe basil", "Thin with garlic and oregano and standard", ... })

Exercise 2.63:

Let A,B,C,D,E be the respective events that the PC is in Adult bedroom, Child bedroom, Other bedroom, Office or den, Other rooms, then according to given condition, we have:

$$P(A) = 0.03 ; P(B) = 0.15 ; P(C) = 0.14 ; P(D) = 0.40 ; P(E) = 0.28$$

(a) For the event that it must be in the bedroom, the PC has to be in the Adult bedroom or CHild bedroom or the Other room (OR means $\cup$) i.e. we have: $P(A \cup B \cup C) = P(A) + P(B) + P(C) = 0.03 + 0.15 + 0.14 = 0.32$

(b) For the event that it must not be in the bedroom, we have to apply complimentary rule to the above event i.e. $P((A \cup B \cup C)') = 1 - P(A \cup B \cup C) = 1 - 0.32 = 0.68$

(c) Since office or den has the highest probability of having a PC, we would expect it to be there.

Exercise 2.64:

Let A and B be the respective events that the component survives more than 6000 hours and no longer than 4000 hours, then according to given condition:

$$P(A) = 0.42 ; P(B) = 0.04$$

(a) For the event that life of components is less than or equal to 6000 hours, it should not be more than 6000 hours, hence, this is a complimentary event of A i.e. $P(A') = 1 - P(A) = 1 - 0.42 = 0.58$

(b) For the event that life of component is greater than 4000 hours, it should not be less than 4000 hours, hence, this is complimentary event of B i.e. $P(B') = 1 - P(B) = 1 - 0.04 = 0.96$

Exercise 2.65:

Now based on the new condition, we have: $P(A) = 0.20 ; P(B) = 0.35$

(a) For the event that component should not fail test, it is the complimentary event of A i.e. $P(A') = 1 - P(A) = 1 - 0.20 = 0.80$

(b) For the event that neither strain is displayed nor it fails the test, we have this as a complimentary event of the event that it either displays strain or fails the test i.e. $(A \cup B)$. Hence, we have:

$$P((A \cup B)') = 1 - P(A \cup B) = 1 - P(A) - P(B) = 1 - 0.2 - 0.35 = 0.45$$

(c) We have already computed the probability of this event i.e. $P(A \cup B) = 0.2 + 0.35 = 0.55$

Exercise 2.66:

According to given condition, we have the following table for the probabilities:

Shift	Unsafe Conditions	Human Error
Day	0.05	0.32
Evening	0.06	0.25
Graveyard	0.02	0.30

(a) From the above table, adding both the probabilities for Unsafe conditions and Human error, we get the probability for accidents occurred on graveyard: $P(A) = 0.02 + 0.30 = 0.32 = 32\%$

(b) From the above table, adding all three probabilities for Day, Evening and Graveyard shifts for the Human error, we get the probability for accidents that occurred due to Human error: $P(B) = 0.32 + 0.25 + 0.30 = 0.87$

(c) From the above table, adding all three probabilities for Day, Evening and Graveyard shifts for the Unsafe conditions, we get the probability for accidents that occurred due to Unsafe conditions: $P(C) = 0.05 + 0.06 + 0.02 = 0.13$

(d) From the above table, for the case of accident occurring on either evening or graveyard shift, we have:

$$P(D) = P(\text{Evening} \cup \text{Graveyard}) = P(\text{Evening}) + P(\text{Graveyard}) = (0.06 + 0.25) + (0.02 + 0.30) = 0.31 + 0.32 = 0.63$$

Exercise 2.67:

(a) For the event that no more than 4 cars are serviced, the mechanic can service either 3 cars or 4 cars. Thus, using additive rule of probability, we have: $P(A) = P(3) + P(4) = 0.12 + 0.19 = 0.31$

(b) For the event that mechanic serves no more than 8 cars, we can compute the complimentary rule on the probability for him serving 8 or more cars as: $P(B) = 1 - P(8 \text{ or more}) = 1 - 0.07 = 0.93$

(c) For the event that he will service either 3 or 4 cars will be the same as in case(a) i.e. $P(C) = P(A) = 0.31$

Exercise 2.68:

(a) Let A be the event that the oven gets purchased by atmost 2 individuals, then according to given condition, we have: $P(A) = 0.4$. Now, for the case of three individuals, the event will be the complimentary of the given as other values have not been specified: $P(A') = 1 - P(A) = 1 - 0.4 = 0.6$

(b) Let B and C be the respective events that all six buy the electric oven and gas oven. According to given condition, we have: $P(B) = 0.007$ and $P(C) = 0.104$. For atleast one sell of both products, we would have to compute the probability of the complimentary event for the individuals buying all electric ovens or gas ovens i.e.

$$P((B \cup C)') = 1 - P(B \cup C) = 1 - P(B) - P(C) = 1 - 0.007 - 0.104 = 0.889$$

Exercise 2.69:

Based on the given condition, we have: $P(A) = 0.990$, $P(B) = 0.001$. Now

(a) Since events A,B and C form the sample space, we have: $P(C) = 1 - P(A) - P(B) = 1 - 0.990 - 0.001 = 0.009$ (b) For the machine to not underfill, we have the complimentary event for event B i.e. $P(B') = 1 - P(B) = 1 - 0.001 = 0.999$ (c) For the machine to either overfill or underfill, we have to find probability of the event $(B \cup C)$ i.e. we have: $P(B \cup C) = P(B) + P(C) = 0.001 + 0.009 = 0.010$

Exercise 2.70:

According to the new condition, we have: Product Price = 4.00 ; Purchase Price: 4.50 ; Number of boxes = 50

(a) The weekly profit in case of no defective boxes can be given by:

$$\begin{aligned} \text{Total Profit} &= (\text{Profit per box}) * (\text{Number of boxes}) = (\text{Purchase Price} - \text{Product Price}) * 50,000 \\ &= (4.50 - 4.00) * 50,000 = \$25,000 \end{aligned}$$

(b) The loss in profit due to underfilling can be determined by finding the expected number of unfilled boxes and then multiplying it with the profit per box, which would give us how much profit is not being earned for those boxes. Since, the probability of getting unfilled box is 0.001 (as per last exercise), we have:

$$\text{Loss in Profit} = (\text{Profit per Box}) * (\text{Number of unfilled boxes}) = (4.50 * 4.00) * (50,000 * 0.001) \\ = 0.5 * 50 = \$25$$

Exercise 2.71:

According to the new condition, let A and B be the events that the product meets weight requirements and too light respectively, then $P(A) = 0.95$ and $P(B) = 0.002$. Moreover, we have: Product Price = 20.00 ; Purchase Price = 25.00.

(a) For the event that the product is too heavy, it does not have to meet requirements or be light i.e. $P(C) = 1 - P(A) - P(B) = 1 - 0.95 - 0.002 = 0.048$

(b) For the case of 10,000 boxes which meet weight requirement, we have total profit as:

$$\text{Total Profit} = (\text{Profit per box}) * (\text{Number of boxes}) = (25-20) * 10,000 = \$50,000.$$

(c) In case of event where all rejected packages are returned, we know that packages which are either heavy or too light would be rejected. The probability of having such packages forms the event $(A \cup B)$. Now, if we determine the expected number of rejected boxes and then multiply it with the gained profit, we will get the value as how much profit is being lost due to these rejected boxes. Since, the probability of getting rejected box is $P(B) + P(C) = 0.002 + 0.048 = 0.05$, we have:

$$\text{Loss in Profit} = (\text{Profit per Box}) * (\text{Number of rejected boxes}) = (25-20) * (10,000 * 0.05) = \$2,500.$$

Chapter 2 Section 4

Examples:

Example 2.34:

For given condition, we have: $P(A) = 0.82$, $P(D) = 0.83$, $P(A \cap D) = 0.78$

(a) For the plane to arrive on time such that it departed on time, means that we need to find the conditional probability for event A such that D has already occurred, which can be found using:

$$P(A|D) = P(A \cap D)/P(D) = 0.78/0.83 = 0.94$$

(b) For the plane to depart on time such that it arrived on time, means that we need to find the conditional probability for event D such that A has already occurred, which can be found using:

$$P(D|A) = P(A \cap D)/P(A) = 0.78/0.82 = 0.95$$

Example 2.35:

Let A and B be the events such that the strip fails the length and texture tests respectively, then according to given condition, we have: $P(A) = 10\% = 0.1$; $P(B) = 5\% = 0.05$; $P(A \cap B) = 0.8\% = 0.008$

Now, for the event that the strip fails the texture test knowing that it has failed the length test, can be found using conditional probability formula as:

$$P(B|A) = P(A \cap B)/P(A) = 0.008/0.1 = 0.08$$

Example 2.36:

Since for the given case, we are told that the fuse picked from the box is removed from it, this means that the events of selecting the defected ones are independent of each other. Now, since the events are independent, therefore, we can find the total probability of the two events by using the multiplication rule as:

$P(A) = (5/20)(4/19) = 20/380 = 1/19$ (Since there were 5 available defective fuses, we had to pick one, which after being removed from the box left us with 4 defective ones and a total of 19 fuses).

Example 2.37:

Let W_1 and B_1 be the event that a white and black ball were selected from first bag. Let W_2 and B_2 be the event that a white and black ball were selected from second bag. Now, we can express this case as the following expression:

$$(W_1 \text{ or } B_1) \text{ and } (B_2) = (W_1 \text{ and } B_2) \text{ or } (B_1 \text{ and } B_2) = (W_1 \cap B_2) \cup (B_1 \cap B_2)$$

Applying probability, we have:

$$P((W_1 \cap B_2) \cup (B_1 \cap B_2)) = P(W_1 \cap B_2) + P(B_1 \cap B_2) = P(W_1)P(B_2|W_1) + P(B_1)P(B_2|B_1) = (4/7)(5/9) + (3/7)(6/9) = 20/63 + 18/63 = 38/63$$

(In this case, when we took a white or black ball from first bag, the total number of balls in the bag reduced by 1. Moreover, when this ball was put in the second bag, it increased the total number of balls in second bag by 1, depending on which ball we drew, the number of white or black balls in second bag were also increased by 1).

Example 2.38:

Let A be the event that fire engine is available and B be the event that the ambulance is available, then we have: $P(A) = 0.98$ and $P(B) = 0.92$. For the required event, we have to find out the probability of the two available at the same time i.e. using multiplication rule, we have:

$$P(A \cap B) = P(A) * P(B) = 0.98 * 0.92 = 0.9016 \text{ (Events are independent of each other)}$$

Example 2.39:

For the given system, we have: $P(A) = 0.9$; $P(B) = 0.9$; $P(C) = 0.8$; $P(D) = 0.8$

(a) For the system to work, the input entering from first line should reach the last line. Through this path, A and B are serially connected to each other which is then connected to a parallel component containing C and D parts. Thus, for the system to work, the input must pass from A,B and (C or D). Using probability expressions, we have:

$$P(A \cap B \cap (C \cup D)) = P(A) * P(B) * P(C \cup D)$$

Using addition rule for mutually inclusive events (this is because they are sharing the same input), we have:

$$\Rightarrow P(A) * P(B) * (P(C) + P(D) - P(C \cap D)) = 0.9 * 0.9 * (0.8 + 0.8 - (0.8)(0.8)) = 0.9 * 0.9 * 0.96 = 0.7776 \text{ (We have } P(C \cap D) = P(C) * P(D) \text{ because of the independence of each component from others)}$$

(b) For the event that the system is working but the component C is not, we will have the following expression:

$$P(A \cap B \cap C' \cap D) = P(A) * P(B) * P(C') * P(D) = 0.9 * 0.9 * (1-0.8) * (0.8) = 0.1296$$

The above probability is for the system if the C component is not working. Now, for the system to work such that C is not working we have:

$$P(E) = P(C \text{ is not working})/P(\text{System is working}) = 0.1296/0.7776 = 0.167$$

Example 2.40:

For the given events, we have:

A_1 : First card is a red ace $\Rightarrow P(A_1) = 2/52$ (There are 2 red aces in a deck)

A_2 : Second card is a 10 or a jack $\Rightarrow P(A_2|A_1) = 8/51$ (There are 4 10s and 4 jacks in a deck. Since first card was drawn total cards are now 51).

A_3 : Third card is greater than 3 but less than 7 $\Rightarrow P(A_3|A_1 \cap A_2) = 12/50$ (The cards can be 4s, 5s or 6s, all of which are 4 in number in a deck. Since the first and second cards were drawn, total cards are now 50)

Now, for the required event, we have:

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) * P(A_2|A_1) * P(A_3|A_1 \cap A_2) = 2/52 * 8/51 * 12/50 = 8/5525$$

Exercises:

Exercise 2.73:

(a) The event that a convict committed armed robbery, who pushed dope. **(b)** The event that a convict who had committed armed robbery, did not push dope **(c)** The event that a convict who did not push dope, did not commit an armed robbery.

Exercise 2.74:

For the given event, total number of students who have earned A are: $O(S) = 3 + 10 + 5 = 18$. Moreover, we have 10 such seniors which have an A i.e. $O(A) = 10$. Now, we have using probability rule:

$$P(A) = O(A)/O(S) = 10/18 = 5/9$$

Exercise 2.75:

Consider the following table:

Education	Male	Female	
Elementary	38	45	83
Secondary	28	50	78
College	22	17	39
	88	112	200

(a) From the above table, we have: $P(\text{Male}|\text{Secondary}) = P(\text{Male} \cap \text{Secondary})/P(\text{Secondary}) = (28/200) / (78/200) = 28/78 = 14/39$

(b) From the above table, we have: $P(\text{College}'|\text{Female}) = P(\text{College}' \cap \text{Female})/P(\text{Female}) = (95/200) / (112/200) = 95/112$

(The operation $\cap$ in both parts means the value that is present at the crossing of that row and column e.g. 28 is the value that lines at the intersection of the row "Secondary" and column "Male" so we got $P(\text{Male} \cap \text{Secondary})$ as $28/200$. In the second part, the condition stated values where gender is female and not in college, so we added the row values for all where gender was "Female" except for the "College" row and thus, we got $P(\text{College}' \cap \text{Female}) = (45 + 50)/200 = 95/200$)

Exercise 2.76:

Consider the following table:

EventRep	NonSmokers	Moderate Smokers	Heavy Smokers	
H	21	36	30	87
NH	48	26	19	93
	69	62	49	180

(a) From the above table, we have $P(H|\text{Heavy Smoker}) = P(H \cap \text{Heavy Smoker})/P(\text{Heavy Smoker}) = (30/200) / (49/200) = 30/49$.

(b) From the above table, we have $P(\text{NonSmoker}|\text{NH}) = P(\text{NonSmoker} \cap \text{NH})/P(\text{NH}) = (48/200) / (93/200) = 48/93 = 16/31$

Exercise 2.77:

Let M be the event that student studied Mathematics, H be the event that he studied History and PY be the event that he studied Psychology, then according to given data:

$$P(M) = 42/100 ; P(PY) = 68/100 ; P(H) = 54/100 ; P(M \cap H) = 22/100 ; P(M \cap PY) = 25/100 ; \\ P(H \cap M' \cap PY') = 7/100 ; P(M \cap PY \cap H) = 10/100 ; P(M' \cap PY' \cap H') = 8/100$$

$$(a) P(M \cap PY \cap H | PY) = P((M \cap PY \cap H) \cap PY) / P(PY) = P(M \cap PY \cap H) / P(PY) = 10/68 = 5/34$$

$$(b) P(M \cap H | PY') = P(M \cap H \cap PY') / P(PY') = (P(M \cap H) - P(M \cap PY \cap H)) / (1 - P(PY)) = ((22 - 10)/100) / (1 - 68/100) = (12/100) / (32/100) = 12/32 = 3/8$$

(in part **(b)**, we have: $P(M \cap H \cap PY') = (P(M \cap H) - P(M \cap PY \cap H))$, because the set $(M \cap H)$ contains all those students who take both Maths and History where as the set $(M \cap PY \cap H)$ contains all those students that are in both Maths and History as well as in Psycology, thus, removing the latter set values from the prior one, we will get all those students which are enrolled in both Mathematics and History but not in Psycology i.e. $P(M \cap H \cap PY')$).

Exercise 2.78:

Let D_1 , D_2 and D_3 be the events that the vaccine is rejected by the first, second and third department respectively, then according to given condition: $P(D_1) = 0.10$; $P(D_2) = 0.08$; $P(D_3) = 0.12$

(a) We have to find the probability for an event that it is not rejected by first department but gets by the second department. Since, the checking is sequential, it can be represented using the AND keyword ($\cap$ operation) as:

$$P(D_1' \cap D_2) = P(D_1') * P(D_2) = (1 - P(D_1)) * 0.08 = (1 - 0.10)(0.08) = 0.072 \text{ (Direct formula can be used because events are independent of each other)}$$

(b) We have to find the probability for an event that it is rejected by the third department, meaning that it was not rejected by the first two. Since, the checking is sequential, it can be represented using the AND keyword ($\cap$ operation) as:

$$P(D_1' \cap D_2' \cap D_3) = P(D_1') * P(D_2') * P(D_3) = (1 - 0.10)(1 - 0.08)(0.12) = 0.099$$

Exercise 2.79:

(a) From the given table, the value intersecting the "Female" column and "Nothing" Row will give us the probability of the female who sleep in nude i.e. $P(A) = 0.018$

(b) From the given table, adding all values in the "Male" column will give us the probability that the traveller is a male i.e. $P(B) = 0.220 + 0.002 + 0.160 + 0.102 + 0.046 + 0.084 = 0.614$

(c) From the given table, the probability that a traveller sleeps in pajamas, such that he is a male can be given as:

$$P(C) = P(\text{Pajamas} | \text{Male}) = P(\text{Pajamas} \cap \text{Male}) / P(\text{Male}) = 0.102 / 0.614 = 0.166$$

(d) From the given table, the probability that a traveller is a male, such that he sleeps in pajamas or T-shirt can be given as:

$$P(D) = P(\text{Male} | \text{Pajamas} \cup \text{T-shirt}) = P(\text{Male} \cap (\text{Pajamas} \cup \text{T-shirt})) / P(\text{Pajamas} \cup \text{T-shirt}) = P((\text{Male} \cap \text{Pajamas}) \cup P(\text{Male} \cap \text{T-Shirt})) / (P(\text{Pajamas}) + P(\text{T-Shirts})) = (0.102 + 0.046) / (0.175 + 0.134) = 0.148/0.309 = 0.479$$

Excercise 2.80:

Let O be the event that oil needs to be changed and F be the event that oil filter needs to be changed, then according to given condition: $P(O) = 0.25$; $P(F) = 0.40$; $P(O \cap F) = 0.14$

(a) The event that oil filter needs to be changed knowing that oil has been changed is given by:

$$P(F|O) = P(F \cap O) / P(O) = 0.14/0.25 = 0.56$$

(b) The event that oil has to be changed such that oil filter is needed to change, we have the probability of event given by:

$$P(O|F) = P(O \cap F) / P(F) = 0.14/0.40 = 0.35$$

Excercise 2.81:

Let M be the event that the man watches a show and W be the event that his wife watches the show, then according to given condition: $P(M) = 0.4$; $P(W) = 0.5$; $P(M|W) = 0.7$

(a) For the married couple to watch a show, they both have to watch it at the same time, hence we have:

$$P(M \cap W) = P(W) * P(M|W) = 0.5 * 0.7 = 0.35$$

(b) For the wife to watch the show such that her husband does, we have:

$$P(W|M) = P(W \cap M) / P(M) = 0.35/0.4 = 0.875$$

(c) For at least one member of the couple to watch a show, either the man should watch it or her wife should i.e.

$$P(M \cup W) = P(M) + P(W) - P(M \cap W) = 0.4 + 0.5 - 0.35 = 0.55$$

Excercise 2.82:

Let H be the event that husband votes and W be the event that his wife votes, then according to given condition: $P(H) = 0.21$; $P(W) = 0.28$; $P(H \cap W) = 0.15$

(a) For at least one member to give a vote, either the husband should give a vote or his wife should i.e.

$$P(H \cup W) = P(H) + P(W) - P(H \cap W) = 0.21 + 0.28 - 0.15 = 0.34$$

(b) For a wife to give vote such that her husband does, we have the probability as:

$$P(W|H) = P(W \cap H) / P(H) = 0.15/0.21 = 0.714$$

(c) For a husband to vote such that his wife does not, we have the probability as:

$$P(H|W') = P(H \cap W') / P(W') = (P(H) - P(H \cap W)) / (1 - P(W)) = (0.21 - 0.15) / (1 - 0.28) = 0.06 / 0.72 = 0.083$$

(We have $P(H \cap W') = P(H) - P(H \cap W)$ because $P(H)$ represents the total probability that a husband will cast a vote whereas the $P(H \cap W)$ represents the probability that a husband will cast a vote with his wife, hence removing the later from the prior will give us the probability that a husband will cast a vote without his wife)

Excercise 2.83:

Let CA be the event that vehicle has Canadian license plate and CM be the event that vehicle is a Camper, then according to given condition, we have: $P(CA) = 0.12$; $P(CM) = 0.28$; $P(CA \cap CM) = 0.09$

(a) The event that a camper entering has a canadian license plate can be given by:

$$P(CA|CM) = P(CA \cap CM) / P(CM) = 0.09 / 0.28 = 0.32$$

(b) The event that a vehicle with canadian license entering is a camper can be given by:

$$P(CM|CA) = P(CM \cap CA) / P(CA) = 0.09 / 0.12 = 0.75$$

(c) The event that a vehicle is neither a camper nor has a canadian license plate can be given by:

$$P(CA' \cup CM') = 1 - P((CA' \cup CM')') = 1 - P(CA \cap CM) = 1 - 0.09 = 0.91$$

Excercise 2.84:

Let H be the event that the head is home and G be the event that goods are brought, then according to given condition: $P(H) = 0.4$; $P(G|H) = 0.3$

For required event, we have the case if the head is home and goods are brought, which can be given by:

$$P(H \cap G) = P(H) * P(G|H) = 0.4 * 0.3 = 0.12$$

Excercise 2.85:

Let D be the event that doctor diagnoses successfully and L be the event that lawsuit is filed against him, then according to given condition: $P(D) = 0.7$; $P(L|D') = 0.9$

For the required event, we have the case if doctor makes an incorrect diagnosis and lawsuit is filed against him, which can be given by:

$$P(D' \cap L) = P(D') * P(L|D') = (1 - 0.7) * 0.9 = 0.27$$

Exercise 2.86:

Let A be the event that four years of college were completed in 1970, W be the event that a female completed four years of college. Let B be the event that four years of college were completed in 1990, then according to given condition: $P(A) = 0.11$; $P(W|A) = 0.43$; $P(B) = 0.22$; $P(W|B) = 0.53$

(a) For the event that the person is a woman such that she completed four years in 1970, we have the probability as: $P(W|A) = 0.43$

(b) For the event that a woman completed her four years of college in 1990, we have the probability as:

$$P(W \cap B) = P(W|B) * P(B) = 0.53 * 0.22 = 0.1166$$

(c) For the event that a man had not finished his college in 1990, we have the probability as:

$$P(W' \cap B') = P(W'|B') * P(B') = (1 - P(W|B)) * (1 - P(B)) = (1 - 0.47) * (1 - 0.22) = 0.53 * 0.78 = 0.4134$$

(Explanation for part (c): We are asked to find the probability of men that have not completed college in four years in 1990 i.e. we have the AND for the events "Men who have not completed college" which is equivalent to the event W' and "Not completed college in the year 1990" which is equivalent to the event B' . We know that both of them are independent events and can be given as the conditional formula rule as: $P(W'|B') * P(B')$. Now, the event $P(B')$ can be easily computed using the complimentary rule, however, for $P(W'|B')$ we have to do some computation. We know that $P(W|B)$ represents all those men who have completed their four years in college in 1990, hence, the complimentary event of this would be those men who have not completed their four years in 1990 which in turn is the event $P(W'|B')$. Finally, we multiply the obtained probabilities to gain the final result)

Exercise 2.87:

Let A be the event that the house is unlocked and B be the event that a master key is selected, then according to given condition. we have: $P(A) = 0.40$

For $P(B)$, we need to find the probability of selecting 3 master keys from a total of 8, such that 1 opens a house while other 2 do not. Hence, we can use the combination formula as:

$$P(B) = P(\text{Select 1 correct key out of 1}) * P(\text{select 2 incorrect keys out of 7}) / P(\text{Select 3 keys out of 8}) = ({}^1C_1 * {}^7C_2) / ({}^8C_3) = (1 * 21)/56 = 21/56 = 0.375$$

Now, for the event that a agent enters a specific house, there are two possibilities i.e. either he gets into an unlocked house with out any keys or he gets into a locked house using a correct key, hence, we get the probability for the event as:

$$P(A \cup (A' \cap B)) = P(A) + P(A' \cap B) = P(A) + (1 - P(A)) * P(B) = 0.40 + (0.60)(0.375) = 0.625 = 5/8$$

Exercise 2.88:

Let A,B,C,D be the events that the CD fails the first, second, third and fourth test respectively, then according to given condition, we have: $P(A) = 0.01$; $P(B) = 0.03$; $P(C) = 0.02$; $P(D) = 0.01$.

(a) For a CD to fail any of the tests, it can either fail first, second, third or fourth test. Hence, we have:

$$P(A \cup B \cup C \cup D) = P(A) + P(B) + P(C) + P(D) = 0.01 + 0.03 + 0.02 + 0.01 = 0.07$$

(b) For a CD to fail the second and third tests, we have the probability as:

$$P(B \cup C) = P(B) + P(C) = 0.03 + 0.02 = 0.05$$

(c) For a sample of 100 CDs, the total probability of the CDs failing any of the tests is 0.07. Hence, the total CDs expected to fail are: Expected = $100 * 0.07 = 7$

(d) Given that the CD is defective, the probability that it was tested can be given by the fact that every four disks are checked at a time i.e. $1/4 = 0.25$

Exercise 2.89:

Let A, B be the events representing that first and second fire engines are available when needed, then we have according to given condition: $P(A) = 0.96$ and $P(B) = 0.96$.

(a) For both the fire engines to be unavailable nor needed, we have the probability as:

$$P(A' \cap B') = P(A') * P(B') = (1 - 0.96) * (1 - 0.96) = 0.04 * 0.04 = 0.0016$$

(b) For a fire engine to be available when needed, either the first one should be available or the second one should be available, we have the probability as:

$$P(A \cup B) = P((A' \cap B')') = 1 - P(A' \cap B') = 1 - 0.0016 = 0.9984$$

Exercise 2.90:

Let A be the event that the river is polluted, B be the event that a sample of water tested detects pollution and C be the event that fishing is permitted, then according to given condition:

$$P(A) = 0.3 ; P(B|A) = 0.75 ; P(B|A') = 0.20 ; P(C|A \cap B) = 0.20 ; P(C|A' \cap B) = 0.15 ; P(C|A \cap B') = 0.80 ; P(C|A' \cap B') = 0.9$$

$$(a) P(A \cap B \cap C) = P(A) * P(B|A) * P(C|A \cap B) = 0.3 * 0.75 * 0.20 = 0.045$$

$$(b) P(B' \cap C) = P((A \cap B' \cap C) \cup P(A' \cap B' \cap C)) = P(A) * P(B'|A) * P(C|A \cap B') + P(A') * P(B'|A') * P(C|A' \cap B') = (0.3 * (1 - 0.75) * 0.8) + ((1 - 0.3) * (1 - 0.20) * 0.9) = 0.06 + 0.504 = 0.564$$

$$(c) P(C) = P(B \cap C) \cup P(B' \cap C) = P((A \cap B \cap C) \cup P(A' \cap B \cap C)) + P(B' \cap C) = P(A) * P(B|A) * P(C|A \cap B) + P(A') * P(B|A') * P(C|A' \cap B) + P(B' \cap C) = (0.3 * 0.75 * 0.2) + ((1 - 0.3) * 0.20 * 0.15) + 0.564 = 0.045 + 0.021 + 0.564 = 0.63$$

(d) The event that the river is polluted given that fish is permitted and sample does not give pollution can be given as:

$$P(A|B' \cap C') = P(A \cap B' \cap C) / P(B' \cap C) = 0.06/0.564 = 0.106 \text{ (Values taken from part (b))}$$

Excercise 2.91:

(a) Let Q_1, Q_2, Q_3, Q_4 be the quarts of milk selected, then we have:

$$P(Q_1 \cap Q_2 \cap Q_3 \cap Q_4) = P(Q_1) * P(Q_2|Q_1) * P(Q_3|Q_1 \cap Q_2) * P(Q_4|Q_1 \cap Q_2 \cap Q_3) = (15/20)(14/19)(13/18)(12/17) = 91/323$$

(b) The selection of four good quarts of milk can be obtained using the combination rule, we have:

$$P(\text{Select 4 good quarts out of 20}) = P(\text{Select 4 quarts out of 15}) / P(\text{Select 4 quarts out of 20}) = {}^{15}C_4 / {}^{20}C_4 = 1365/4845 = 91/323$$

Excercise 2.92:

For the given figure, we have for the system to work:

$$P(A \cap (B \cup C) \cap D) = P(A) * P(B \cup C) * P(D) = P(A) * (P(B) + P(C) - P(B \cap C)) * P(D) = 0.95 * (0.7 + 0.8 - (0.7)(0.8)) * 0.9 = 0.95 * 0.94 * 0.9 = 0.8037 \text{ (As explained in example 2.39)}$$

Excercise 2.93:

For the given figure, we have:

(a) For the event that the entire system works, we have the probability as:

$$P((A \cap B) \cup (C \cap D \cap E)) = P(A \cap B) + P(C \cap D \cap E) - P((A \cap B) \cap (C \cap D \cap E)) = P(A)P(B) + P(C)P(D)P(E) - P(A)P(B)P(C)P(D)P(E) = (0.7 * 0.7) + (0.8 * 0.8 * 0.8) - (0.7 * 0.7 * 0.8 * 0.8 * 0.8) = 0.49 + 0.512 - 0.2508 = 0.7512 \text{ (The explanation is the same as in excercise 2.92 and example 2.39)}$$

(b) For the event that the entire system works such that A is not working can be given by:

$$P(A'|S) = P(A' \cap C \cap D \cap E) / P(S) = P(A')P(C)P(D)P(E) / P(S) = (1 - 0.7)(0.8)(0.8)(0.8) / (0.7512) = 0.1536/0.7512 = 0.2044$$

(Here we omitted the event B in the numerator as it is clear from the figure that if the part A of the system does not work, the input never reaches the component B and it is as if the

Chapter 2 Section 5

Examples

Example 2.41:

Let D be the event that the machine is defective, B_1 be the event that product is made by machine B_1 , B_2 be the event that product is made by machine B_2 and B_3 be the event that product is made by machine B_3 , then according to given condition, we have:

$$P(B_1) = 0.30 ; P(B_2) = 0.45 ; P(B_3) = 0.25 ; P(A|B_1) = 0.02 ; P(A|B_2) = 0.03 ; P(A|B_3) = 0.02$$

For the event that the product is defective, we know that the product is produced using three kinds of machine classes i.e B_1 , B_2 and B_3 . Hence, to find the total probability of defective product, the condition meets that for the total probability rule, hence, we have:

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3) = (0.3)(0.02) + (0.45)(0.03) + (0.25)(0.02) = 0.006 + 0.0135 + 0.005 = 0.0245$$

Example 2.42:

Now, in cases like these, where we have to find the probability of a particular partition based on some condition, we apply the Bayes rule (as B_3 forms one of the partition for the manufacturing of the product) as:

$$P(B_3|A) = P(B_3)P(A|B_3) / P(A) = (0.25)(0.02) / (0.0245) = 0.005/0.0245 = 0.204$$

Example 2.43:

Let D be the event that the procedure is defective, P_1 be the event that plan 1 is selected, P_2 be the event that plan 2 is selected and P_3 be the event that plan 3 is selected, then according to given condition:

$$P(P_1) = 0.3 ; P(P_2) = 0.2 ; P(P_3) = 0.5 ; P(D|P_1) = 0.01 ; P(D|P_2) = 0.03 ; P(D|P_3) = 0.02$$

To find the most likely plan that caused the defect, we have to compute the probabilities for each of the plan used separately and then compare them to get our result. First, finding the total probability of the defect rate i.e.

$$P(D) = P(P_1)P(D|P_1) + P(P_2)P(D|P_2) + P(P_3)P(D|P_3) = (0.3)(0.01) + (0.2)(0.03) + (0.5)(0.02) = 0.003 + 0.006 + 0.01 = 0.019$$

Now, for the individual probabilities caused by the defective components, we have:

$$P(P_1|D) = P(P_1 \cap D) / P(D) = (P(P_1) * P(D|P_1)) / P(D) = (0.3 * 0.01)/0.019 = 0.003/0.019 = 0.158$$

$$P(P_2|D) = P(P_2 \cap D) / P(D) = (P(P_2) * P(D|P_2)) / P(D) = (0.2 * 0.03)/0.019 = 0.006/0.019 = 0.318$$

$$P(P_3|D) = P(P_3 \cap D) / P(D) = (P(P_3) * P(D|P_3)) / P(D) = (0.5 * 0.02)/0.019 = 0.01/0.019 = 0.053$$

From the above situation, we have that the product is defective most likely because of defect in plan 3.

Exercices:

Exercise 2.95:

Let C be the event that an adult has cancer and D be the event that doctor diagnoses a patient as having cancer, then according to given condition, we have:

$$P(C) = 0.05 ; P(D|C) = 0.78 ; P(D|C') = 0.06$$

For the event that a doctor diagnoses an adult over 40 years having cancer can be given by:

$$P(D) = P(C \cap D) + P(C' \cap D) = P(C)P(D|C) + P(C')P(D|C') = (0.05)(0.78) + (0.95)(0.06) = 0.039 + 0.057 = 0.096$$

(A confusion that may arise during this exercise is that why did we declare the event D as the event that "doctor diagnoses a patient as having cancer" instead of denoting it with the event that "doctor diagnoses the cancer correctly". It must be known that the doctor specified in this case knows how to diagnose cancer only, regardless of the illness. Hence, even if the disease is not cancer, he will just carry out the diagnose treatment for cancer, alternatively meaning that he incorrectly diagnoses the person.)

Exercise 2.96:

Let S be the event that a person receives a speeding ticket, L_1 be the event that person passes through location L_1 , L_2 be the event that person passes through location L_2 , L_3 be the event that person passes through location L_3 and L_4 be the event that person passes through location L_4 , then according to given condition:

$$P(L_1) = 0.4 ; P(L_2) = 0.3 ; P(L_3) = 0.2 ; P(L_4) = 0.3 ; P(S|L_1) = 0.2 ; P(S|L_2) = 0.1 ; P(S|L_3) = 0.5 ; P(S|L_4) = 0.2$$

Now, for the event that a person will receive a speeding ticket when off to work, we have:

$$P(S) = P(S \cap L_1) + P(S \cap L_2) + P(S \cap L_3) + P(S \cap L_4) = P(L_1)P(S|L_1) + P(L_2)P(S|L_2) + P(L_3)P(S|L_3) + P(L_4)P(S|L_4) = (0.4)(0.2) + (0.3)(0.1) + (0.2)(0.5) + (0.3)(0.2) = 0.08 + 0.03 + 0.10 + 0.06 = 0.27$$

Exercise 2.97:

For the event that the person diagnosed as having cancer actually has a cancer, we have using Bayes rule:

$$P(C|D) = (P(C) * P(D|C)) / P(D) = (0.05)(0.78) / (0.096) = 0.039 / 0.096 = 0.406$$

Exercise 2.98:

For the event that the person recieved her speed ticket, the probability of her passing through location L_2 can be given using Bayes rule as:

$$P(L_2|S) = (P(L_2) * P(S|L_2)) / P(S) = (0.3)(0.1) / (0.27) = 0.03/0.27 = 0.111$$

Exercise 2.99:

Let F be the event that a person fails to stamp expiration date, A be the event that John stamps the package, B be the event that Tom stamps the package, C be the event that Jeff stamps the package and D be the event that Pat stamps the package, then according to given condition:

$$P(A) = 0.2 ; P(B) = 0.6 ; P(C) = 0.15 ; P(D) = 0.05 ; P(F|A) = \text{once per 200 packages} = 1/200 = 0.005 ; P(F|B) = \text{once per 100 packages} = 1/100 = 0.01 ; P(F|C) = \text{once per 90 packages} = 1/90 = 0.011 ; P(F|D) = \text{once per 200 packages} = 1/200 = 0.005$$

Now, for the event that customer complains that expiration date is not shown on her packages, the probability that it was inspected by John can be given using Bayes rule as:

$$P(A|F) = P(A \cap F) / P(F) = (P(A) * P(F|A)) / (P(A)P(F|A) + P(B)P(F|B) + P(C)P(F|C) + P(D)P(F|D)) = (0.2 * 0.005) / (0.2 * 0.005 + 0.6 * 0.01 + 0.15 * 0.011 + 0.05 * 0.005) = 0.001 / (0.001 + 0.006 + 0.0016 + 0.00025) = 0.001 / 0.00885 = 0.1129$$

Exercise 2.100:

Consider the table:

	Station	A	B	C
Problems with electricity supplied	2	1	1	4
Computer malfunction	4	3	2	9
Malfunctioning electrical equipment	5	4	2	11
Caused by other human errors	7	7	5	19
	18	15	10	43

Let E be the event that malfunction was caused by Human error then we have using the given table:

$$P(A) = 18/43 ; P(B) = 15/43 ; P(C) = 10/43 ; P(E|A) = 7/18 ; P(E|B) = 7/15 ; P(E|C) = 5/10$$

For the event that malfunction was came from station C such that it was caused by other human errors as:

$$P(C|E) = P(C \cap E) / P(E) = (P(C) * P(E|C)) / (P(A)P(E|A) + P(B)P(E|B) + P(C)P(E|C)) = (10/43 * 5/10) / (18/43 * 7/18 + 15/43 * 7/15 + 10/43 * 5/10) = (5/43) / (7/43 + 7/43 + 5/43) = (5/43) / (19/43) = 5/19 = 0.2632$$

Exercise 2.101:

Let A be the event that a customer buys latex paint then A' will be the event that a customer buys semi-gloss paint and B be the event that a customer purchases rollers, then according to given condition, we have:

$$P(A) = 0.75 ; P(B|A) = 0.6 ; P(B|A') = 0.3$$

For the event that a random buyer purchases a paint and roller such that paint is latex, we have the probability as:

$$P(A|B) = P(A \cap B) / P(B) = (P(A) * P(B|A)) / (P(A)P(B|A) + P(A')P(B|A')) = (0.75 * 0.6) / (0.75 * 0.6 + 0.25 * 0.3) = 0.45 / (0.45 + 0.075) = 0.45 / 0.525 = 0.857$$

Exercise 2.102:

Let A, B and C be the events that the price lies behind the three respective doors. Suppose,

In []:

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